

PHY-275 Fall 2026 Assignment #3 due 10/2/26 9:00pm

You will receive 1 credit upon complete all the tasks.

Task #1: Familiar with the ∇ nabla operation (sometimes we write $\vec{\nabla}$ instead of ∇ , and it's the same operation).

If I have a scalar function $f = x_1^2 + x_2^2$

- 1.1) Find ∇f ,
- 1.2) then graph ∇f on the phase portrait (x_1 vs x_2 plane),
- 1.3) then test if grad (∇f) is a linear operation.

Task #2: Familiar with the ∇ nabla operation, again

If I have a vector $\vec{F} = x\hat{i} + y\hat{j}$, or I can express it as $\begin{bmatrix} x \\ y \end{bmatrix}$:

- 2.1) Graph $\vec{F}$ on the phase portrait (x vs y plane).
- 2.2) Find the div: $\nabla \cdot \vec{F}$

Task #3: Familiar with the ∇ nabla operation, again X2

If I have a vector $\vec{F} = -v\hat{i} + x\hat{j}$, or I can express it as $\begin{bmatrix} -v \\ x \end{bmatrix}$:

- 3.1) Graph $\vec{F}$ on the phase portrait (x vs v plane).
- 3.2) Find the curl: $\nabla \times \vec{F}$

Task #4:

- 4.1) Get the curl of a grad: $\nabla \times \nabla f$, where $f = x_1^2 + x_2^2$
- 4.2) Get the div of a curl: $\nabla \cdot \nabla \times \vec{F}$, where $\vec{F} = -v\hat{i} + x\hat{j}$

To proof your efforts, write down each step and upload them to the Moodle site as a single file.